

Architecture Quality Checklist

Assignment 01 – Cloud Architecture Design

Before submitting your assignment, verify that your work satisfies the following quality checks. Also submit hardcopy of this checklist.

1. Problem Understanding

- ☐ The system purpose is clearly explained.
- ☐ The target users and system environment are identified.
- ☐ Functional requirements describe what the system must do.
- ☐ Non-functional requirements include scalability, security, and availability.
- ☐ Each requirement is connected to an architectural implication.

Quick self-check:

If someone reads only Section 1, can they clearly understand what system is being designed?

2. Delivery Model Design

- ☐ The system is broken into multiple components.
- ☐ Each component is assigned an appropriate delivery model (IaaS / PaaS / SaaS).
- ☐ At least **3-4 components** are analyzed.
- ☐ Each delivery model choice is technically justified.
- ☐ A shared responsibility matrix is included.

Quick self-check:

Did you avoid assigning a single delivery model to the entire system?

3. Deployment Model Decision

- ☐ A deployment model is clearly selected.
- ☐ Justification includes data sensitivity considerations.
- ☐ Governance and compliance factors are discussed.
- ☐ Cost implications are explained.
- ☐ Scalability requirements are considered.

Name: _____

Enrollment: _____

Quick self-check:

Did you explain *why this model is appropriate*, not just what it is?

4. Architecture Diagram Quality

- ☐ Diagram clearly shows system layers.
- ☐ Major components are labeled.
- ☐ Delivery model for components is indicated.
- ☐ Deployment boundaries are shown.
- ☐ Trust boundaries are illustrated.
- ☐ Security components are included.

Quick self-check:

If someone sees only the diagram, can they understand how the system works?

5. Cloud Characteristics Implementation

- ☐ Each cloud characteristic is linked to a specific part of the architecture.
- ☐ Explanation focuses on implementation, not definition.
- ☐ Operational impact is discussed.
- ☐ Architecture supports scalability and reliability.

Quick self-check:

Did you explain *how the architecture enables the characteristic*?

6. Risk and Governance Analysis

- ☐ At least **five risks** are identified.
- ☐ Each risk has a root cause explanation.
- ☐ Each risk includes architectural impact.
- ☐ Each risk includes a mitigation strategy.
- ☐ Responsibility (provider vs consumer) is identified.

Quick self-check:

Are your risks specific to your architecture?

Name: _____

Enrollment: _____

7. Engineering Reasoning

- ☐ Architectural decisions are logically justified.
- ☐ Trade-offs are acknowledged.
- ☐ Cost vs control considerations are discussed.
- ☐ Limitations of the design are recognized.

Quick self-check:

Did you explain **why your architecture makes sense**, not just what it is?

8. Academic Integrity

- ☐ All explanations are written in your own words.
- ☐ No diagrams are copied directly from the internet.
- ☐ External references (if used) are properly cited.

9. Formatting Compliance

- ☐ Report length is **7–8 pages maximum**.
- ☐ Sections follow the provided template.
- ☐ Font and spacing follow assignment instructions.
- ☐ Tables and diagrams are readable.

10. Final Engineering Check

Answer these questions in Yes/No/Not Clear:

- Can this architecture **scale if user traffic increases**?

- Can it **handle failures without major disruption**?

Name: _____

Enrollment: _____

-
- Does it **protect sensitive data properly**?

-
- Does it **balance cost and control effectively**?
-

If the answer to these questions is **no/not clear**, improve your design.

Remarks: *(If there is anything different that you would like to explain about your assignment, explain here)*
